Testing in the Field: Voice Based Interaction for Citizen Reporting in Uganda

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Abstract
In developing countries where Internet is not readily accessible and literacy sometimes low, Interactive Voice Response (IVR) systems have been shown to provide opportunities in a variety of civic activities. In this way, such systems can give the opportunity to more easily influence and participate in public affairs. However the lack of an efficient system of administration can delay delivering the collected reports to the relevant organizations or authorities. This paper presents a study of a citizen reporting system that was developed, implemented and tested in Uganda. This was done in collaboration with the Women of Uganda Network an organization to empower women using ICT. We studied an IVR system called U-Call that allows administrators to easily publish and tag audio reports over the Web (reported by citizen) using low generation mobile devices. We highlight a number of issues, including multiple authentication process that needs to be taken into account for future development. We emphasize the importance of field studies in understanding and designing for this user group and understanding the reality of the technical infrastructure available.


HIV Positive Dad Defiles Daughter

A 27-year-old man has been arrested for allegedly defiling his 10-year-old biological daughter and is being held at Fort Portal police station in Boma.

Geoffrey Asaba a resident of Lyamabwa village in Kisomoro sub county in Kabarole district was arrested on Monday.

"Asaba was coming from a night club and was under the influence of alcohol," Mugga Bakari the regional police spokesperson said.

"He had threatened to harm the victim if she told anyone" Bakari added.

Audio:

Download:

9597E902-5ACE-49B6-A0F2-234A6BC63CA3.m4a.mp3

» Read more

Figure 1: A screenshot of a published report on U-Call Web interface and the link to the related audio file. The case is reported by a citizen (audio file) then it verified, transcribed and published by WOUGNET.

Author Keywords

ICT4D, mobile voice systems, field studies

ACM Classification Keywords

H.5.m. Information interfaces and presentation (e.g., HCI): Miscellaneous.

Introduction

In developing countries, Interactive Voice Response (IVR) applications are increasingly used to collect voice-based reports from citizens. These can be reports of problems with governance or poor delivery of public services (such as education, healthcare, public infrastructure, etc.) [4,6,7,8]. These systems overcome limited access to the Internet and other technical infrastructure, by leveraging the widespread use of simple mobile phones. They provide an opportunity for rural people in developing countries to influence and participate in public affairs and report their problems to the responsible authorities [1]. However, it is not clear how well these are integrated into their actual rural settings, at least beyond studies such as the ones by Patel et al. [8] and Sherwani et al. [10]. To explore the implementation and feasibility of this type of system we focused on one in particular: U-Call, a system designed prior to our first research trip in the summer of 2012 to the Northern districts of Uganda. This research project produced an ethnographical field study of the use of U-Call [2] and reflections on the design process [9]. Using elements of design research we added features to the system and tested them in the field. We encountered several problems that emphasized the importance of implementing systems within an ecologically valid setting. Our study contributes to a better understanding of not only the limitations of the technical infrastructure in rural Uganda, but also to the broader issues encountered when field testing in a developing country.

Previous Work

Voice-based forums and reporting applications in the developing world are not new, and several systems have been developed and tested. Patel et al. created a forum for Indian farmers to exchange information and experiences [8]. In another project De Boer et al. developed RadioMarche, a voice-based Market Information System for forestry products in the African Sahel that allowed farmers to advertise their products on the radio [3]. Mahelaqua et al. [6] explore the use of multimodal voice browsers to navigate the spoken web by low-literate users in rural India.

The U-Call System

This study is based on U-Call, an IVR reporting system designed to facilitate the communication between people in remote rural areas of Uganda and Women of Uganda Network (WOUGNET), a non-profit organization focusing on empowering women [2]. Similar to previous systems it is based on voice communication to allow access to those with low levels of literacy and limited Internet access. The system allows people to make free calls to report problems in their neighborhood or to listen to problems that have already been reported, without the need for an Internet or application capable mobile device. Figure 1 shows a reported case by a citizen, verified and published (text and audio file) by WOUGNET on the U-Call Web interface.

What most of the previous projects in this field, including U-Call, have in common is that they target local people in remote areas to either collect data from

UCAF: Hello administrator! Welcome to the administrative menu. If you want to access to the admin menu, press 1. For the user menu press 2.
User: *(Presses 1)*
UCAF: Now you can moderate the reports. During playback press any key to go to the audio menu.
UCAF: *(Plays the first audio report)*
UCAF: to replay the current audio press 1. To go to the next audio press 2. To publish the report with tag Gulu, press3. Oyam, press 4. Kampala, press 5. Amuru, press 6. Apach, press 7. To go to the main menu press the pound key.
User: *(Presses 6)*
UCAF: The report is published with tag Amuru.
UCAF: *(Plays the next audio report)*

Figure 2: A sample successful interaction with the U-Call Administration Functionality module.

them or give information to them. Many also make the assumption that the organizations or the authorities that will verify and publish the reports have basic access to the Internet, even if they work in the remote areas themselves. Our initial studies of the WOUGNET field representatives who regularly travel to the remote areas of the Northern districts of Uganda, however, showed that this is not necessarily the case [5].

Despite the ongoing effort of the WOUGNET representatives, the lack of an efficient system of administration can cause delays in broadcasting the collected reports. Our initial field study yielded the following example by one of our participants, a WOUGNET representative: *"… a 13 years old student was beaten by her parents and being forced into marriage against her will. Her teacher reported the case to the local community but they delayed to respond, and they couldn't find me to report directly, my phone was not reachable, I was in another city in a conference I think…the report didn't get to us and other authorities until days after. The ceremony was due to take place…"* This particular case illustrates the importance of timeliness regarding the publication and forwarding of citizens' reports.

Currently the U-Call reporting system needs a system administrator (a WOUGNET staff member) to listen to, and verify the stored and unpublished audio-based reports before publishing them. This control helps WOUGNET eliminate invalid or inappropriate reports. However, most of the time WOUGNET members do not have access to a computer, smartphone or Internet connection, especially while they are in remote areas of the country, and these administrative tasks required Internet access. In an attempt to overcome this bottleneck we therefore implemented a voice controlled administrative system to complement the citizen-oriented one already deployed in U-CALL. This system was developed on the basis of the initial field study as well as further investigations carried out into WOUGNET representatives' work structures.

Design and Implementation of UCAF

During the spring 2013 we conducted several Skype meetings with WOUGNET representative who regularly travel to the rural areas in the Northern districts of Uganda. We asked the interviewees to describe their workflow. This included describing: a) how they collect the reports, b) who publishes the reports and how and, c) What problems they face in order to publish a report on their reporting platform. Using the insights from these interviews, we designed and implemented the U-Call Administrative Functionality (UCAF). It extends the IVR to support the U-Call administrators, allowing them to easily publish and tag audio reports when they are without Internet access. Considering the technology limitations on the U-Call system, our UCAF has only one main component: a Mobile Administrator module that adds more administrative features to both the Drupal for Android module and the Reporting System module. This module is able to distinguish administrative users from normal users to give them access to the more than the published information. In order to provide different voice menus for users and admins, the system first identifies whether the caller is an admin by verifying the caller's phone number is that of an administrator before providing access to the admin menu. If the number is registered as belonging to a WOUGNET administrator the Mobile Administrator module will be loaded instead of the Reporting System module (which is accessible to all), which then calls the







Figure 3: Training WOUGNET staff on how to use the system

user back. The admins then have access to unpublished reports instead of published reports (as Figure 2 illustrates) they can also interact as regular users. The administrator can choose different options: a) listen to the report(s) and, b) listen to the report(s) and publish with the relevant tag. Each key on the phone keypad gives different options to the admin. See Figure 2 for a successful interaction of an admin user with UCAF.

Testing of the System

For practical purposes, we ran the first phase of test and evaluation in Sweden, using both mobile phones and landlines to check UCAF with different Swedish network operators. In July of 2013, we traveled to Uganda again and conducted several field tests and system evaluations in Uganda with WOUGNET staff (Figure 3), both in Kampala and in the rural Northern districts (Oyam, Gulu, Amuru) as well as several parishes such as Toro, Ibuje, Tarogali and Aboke where the WOUGNET field representatives work (Figure 4).

Participants

Six participants from WOUGNET staff were selected for their different roles and responsibilities to work with UCAF. Of the six participants, two participants were responsible for the current WOUGNET reporting portal over the Ushahidi[1] Web site. One participant was assigned to follow the reports. Two other participants were interns at WOUGNET and therefore they had limited experience with existing reporting system. The last participant was WOUGNET's field representative who travels between the districts to collect reports from locals, verify them, and publish these reports using the Web portal.

Workshops and Field Studies

During the first two days we trained and tutored our participants, and successfully tested the system in both rural areas and Kampala. Due to mobile network overload problems in Kampala (acknowledged by the local telecommunication service provider, MTN Uganda[2]), which caused poor quality of phone service coverage on the third day of testing, all the tests using mobile phones failed. However, the tests using the landline received the call back successfully, enabling participants to publish and tag the selected reports. Thus we were able to verify that the integration between U-Call and UCAF was working properly. However, the experience with the mobile network overload exposed problems with the infrastructure in the field we had not expected when designing our system.

For the final test, we traveled together with a WOUGNET field representative to the Northern rural districts. During the 5-hour trip from Kampala to the Northern districts, we called the system every 30 minutes to check its behavior in different locations with different network coverage. Our participant made 20 calls in total, all successfully, including random calls to test the system under different network circumstances.

Initial Findings and Further Developments

Our experience from the field studies in Uganda showed that the MTN operator, which is popular and provides an inexpensive Internet connection, does not have adequate network coverage everywhere to support a system such as U-Call. For example in some of the districts like Amuru or Community Taragali in the Ibuje

[1] http://wougnet.org/ushahidi

[2] http://www.mtn.co.ug







Figure 4: Workshop to test and evaluate the system

sub-county, there was no network coverage and we could not make any phone calls (to the system or at all). "*If you stand on top of that hill, you might get some signals. That's the only place we found around here that MTN sims work*", said one of the locals during the field test at Amuru district, while we couldn't make any calls. One way locals get around this problem, which they are aware of, is to have mobile phones that support two SIM-Cards at the same time. When finding one of the operators not supporting a specific location, they switch to the other SIM-Card, which is supported by other operators such as airtel[3]. According to WOUGNET's statement, their staff members follow the same approach. The consequence for our system was that since we just register one mobile number for WOUGNET admins, they would be disconnected from the admin menu when they switch to the other SIM-Card. Our system then needed to support more than one phone number per person.

During the design phase and tutorial sessions we asked WOUGNET participants if they had preferred to input a pin code when the system calls them back to increase security. One participant who had experience with moderating the admins access to the existing reporting system mentioned, *"This additional step will increase the complexity of accessing the service as our staff will need to remember an additional pin code. Consequently if they forget their pin code they will be disconnected from the system until someone changes it in their user profile"*. However such control appeared necessary to WOUGNET, while in an actual case one of the admin's phone got stolen and they had to make several calls to find another admin member who had access to Internet therefore the system web interface, in order to disable the phone number in the user profile to avoid any future unauthorized access. This whole process took about five hours to remove the stolen phone number from the system. After this field incident, WOUGNET members were convinced to reconsider the pin code option. We suggested implementing a fail-secure method by making the pin code optional. We used the classic credential model based on username-password whereas the username is the caller's phone number, which is automatically detected by our system and the password is the pin code, stored in our system by the user and should be entered during the operation.

[3] http://www.africa.airtel.com

Conclusions and Future Work

In this project we explored the further development of an Interactive Voice Response system for citizen reporting where initial field studies had revealed a need for an administrative system using voice response instead of a web interface. After developing an administrative IVR interface to the U-Call system we studied the integration between the two systems in a field study with WOUGNET – the organization that used the system in the field. Our study showed that while any WOUGNET administrators can use the system independent of access to the Internet, it still needed adjustments to real environmental problems such as limited mobile phone coverage in rural areas. Moreover, while pin codes were more secure and proved to be valuable in real world incidents experienced by the WOUGNET staff, they were resisted and needed to be implemented as an option rather than a default.

The system we implemented was clearly working well, at least in the test conditions in Sweden and Kampala, not considering the cellular network problems.

However, not until the actual field studies did we encounter more real-world problems such as the two SIM-Card solution and users' reluctance to memorize and keep pin codes. This highlights that in developing countries, or anywhere technically and culturally removed from the design and development environment, it is important to perform field tests since the infrastructure can be very different and unpredictable compared to what is reported or expected. What seems to be a simple technological solution for developers in the West might not work on the ground, in the local environment of technical and social infrastructure – an issue those entering ICT4D are repeatedly forced to learn.

In terms of future work, we plan to improve the UCAF based on the administrators' needs. Each administrator could, for example, have access to specific reported cases based on the location, topic, date and etc. We also plan to study and develop a suitable dynamic mobile tagging system by studying the administrators' use of UCAF based on frequently used tags by each administrator.

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